

Math 432/532: Introduction to Topology II

HW #4

1. Suppose that L is a simplicial complex and $K_1, K_2 \subset L$ are subcomplexes. Show that $\chi(K_1 \cup K_2) = \chi(K_1) + \chi(K_2) - \chi(K_1 \cap K_2)$.
2. Suppose that K is a simplicial complex, and consider its barycentric subdivision K^1 . It is true in general that $\chi(K^1) = \chi(K)$, but the proof involves more combinatorial bookkeeping than I want to ask you to do on a homework assignment. Instead, I'm just going to ask you to prove it for complexes of dimension at most 2.
 - (i) Prove it when $\dim K = 1$.
 - (ii) Prove it when $\dim K = 2$.
3. Let X be the Klein bottle, and consider the three simple closed curves shown in Figure 7.16 of your book. For each of these curves, determine whether it thickens to a cylinder or a Möbius strip, and describe the surface obtained by a single surgery. (Note that you can answer this question without thinking about triangulations.)